Outlook

FW: Constant Tension Winch

De Bruno Alves <bruno.alves2@technipfmc.com>
Data Qua, 27/05/2026 16:48
Para Daniel Wanderley <daniel.wanderley@technipfmc.com>

2 anexos (143 KB)
winch with link.dat; winch with link.wrk;

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From: Orcina <orcina@orcina.com>
Sent: quarta-feira, maio 27, 2026 13:10
To: Bruno Alves <bruno.alves2@technipfmc.com>
Subject: Re: Constant Tension Winch

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Dear Bruno,

If I have understood your required model set-up, I suspect that one solution would be to use an OrcaFlex link object in combination with a winch. The reason for this is that it is not possible to directly define a max pay-out / haul-in speed for the winch (even when using the "detailed" type).

As you have correctly identified, the winch is able to pay-out / haul-in to maintain a particular defined constant tension. If you use the "simple" winch type, then the specified tension will be maintained exactly throughout the simulation.

However, it sounds as though you would like to apply a small "tolerance", so that the winch will only pay-out if it exceeds the tension plus the tolerance, or haul-in if it falls below the tension minus the tolerance? If so, then I think the detailed winch deadband should allow you to capture this tolerance-type behaviour.

Essentially, the deadband performs exactly as you've described here - as I hope you can see in the following *Figure* from the help page *Theory | Winch theory*:

where

v = rate of payout = $\mathrm{d}l_0/\mathrm{d}t$

d = winch drive deadband

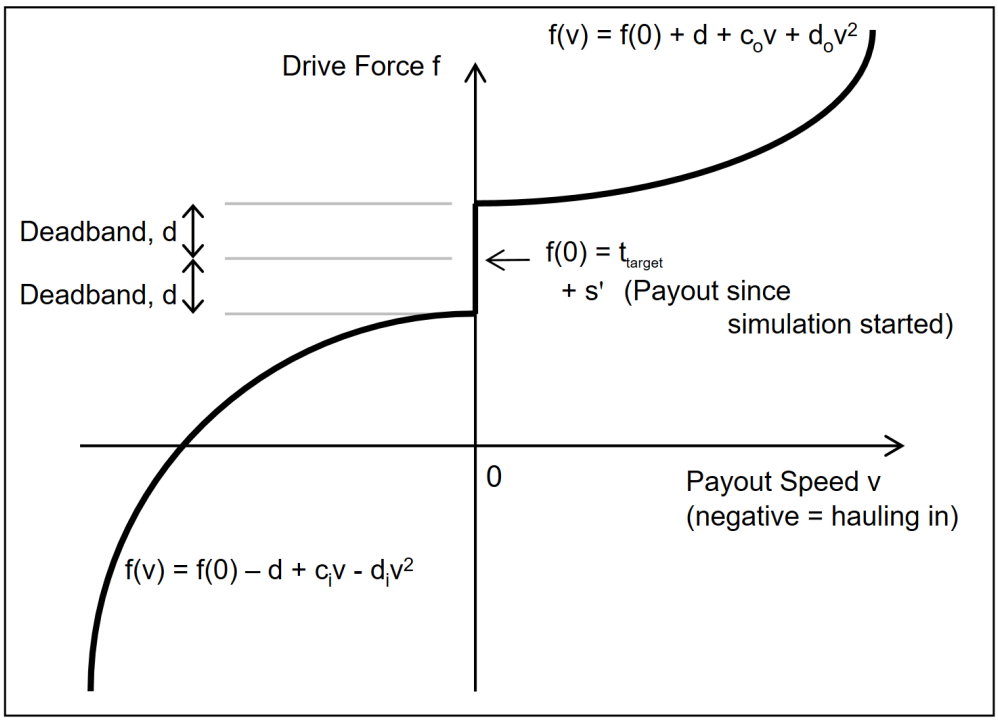
$c_\mathrm{i}, c_\mathrm{o}$ = winch drive damping terms for haul in and pay out

$d_\mathrm{i}, d_\mathrm{o}$ = winch drive drag terms for haul in and pay out

$f(0) = t_{\mathrm{target}} + s\,(l_0 - l_{00})$

s = winch drive stiffness

l_{00} = original value of l_0 at the start of the simulation (set by the static analysis).



As I mentioned above, there is no "built-in" detailed winch option that will limit the pay-out / haul-in rate.

I can see that you've tried to adjust the winch damping and drag data (perhaps to try to limit the pay-out / haul-in rate), but this is not the intended purpose of these data items. As is noted in the help page *Modelling, data and results | Winches | Drive unit*, these data are intended to represent the behaviour of the winch's hydraulic drive system, and affect its behaviour as shown in the figure above.

Probably the easiest way to introduce a limit on the maximum pay-out / haul-in rate is to include a non-linear "spring/damper" link type, which would act in parallel with the detailed winch. I've attached a very simple example file here that tries to show how this could perhaps work.

The idea here is that the link contributes no spring stiffness loads, and only a non-linear variation in damping force with velocity. In the attached file, I've tried to limit the winch velocity to no greater than 0.5m/s. As you'll see, I've then set-up a non-linear tension vs velocity profile for the link, where the tension is zero between velocities of +/-0.5m/s. If the velocity exceeds this range, the tension force then increases significantly, which should hopefully limit the velocity experienced.

If you run the model and take a look at the attached workspace, you'll see time histories of the winch's tenson, length and velocity - as we hope, the velocity has been limited within the required range. If you remove the link, you should see that the velocities will exceed this range.

One comment I would make is that the link sometimes needs to apply a relatively large tension to limit the velocity - in this case, tensions of only ~5kN are required, but far higher values may be necessary for other configurations. I'm not sure whether this will matter for your analysis, but I wanted to make you aware of this behaviour as something to possibly monitor in more detail.

Another point to note is that applying a very large change in tension, over only a very small change in velocity, could make dynamic convergence more difficult. So, you may find that you need to take care when defining the tension vs velocity data, to ensure robust convergence whilst still limiting the velocity in the intended way.

Please feel free to use the attached model to make your own experiments, just to check whether this method could offer you something useful for your analysis.

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As a final thought, I think it would also be possible to replicate the behaviour that you want by perhaps "taking over" control of the winch yourself, through an external function. But it's possible that this level of complexity may not be required here, particularly if you felt that using a link in combination with the winch could offer a solution.

I hope that this helps, but please do get back in touch if there's anything I've mentioned here that you would like to discuss in more detail or if you have any problems when trying to build the model for yourself.

Best Regards,

Sarah

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From: Bruno Alves <bruno.alves2@technipfmc.com>
Sent: 26 May 2026 10:23 PM
To: Orcina <orcina@orcina.com>
Subject: Constant Tension Winch

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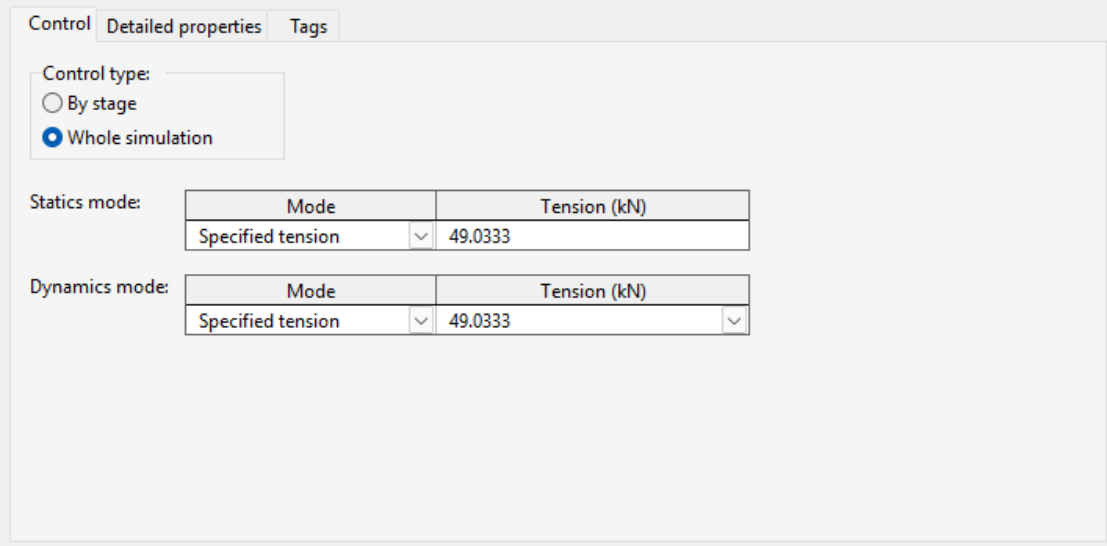
Good afternoon,

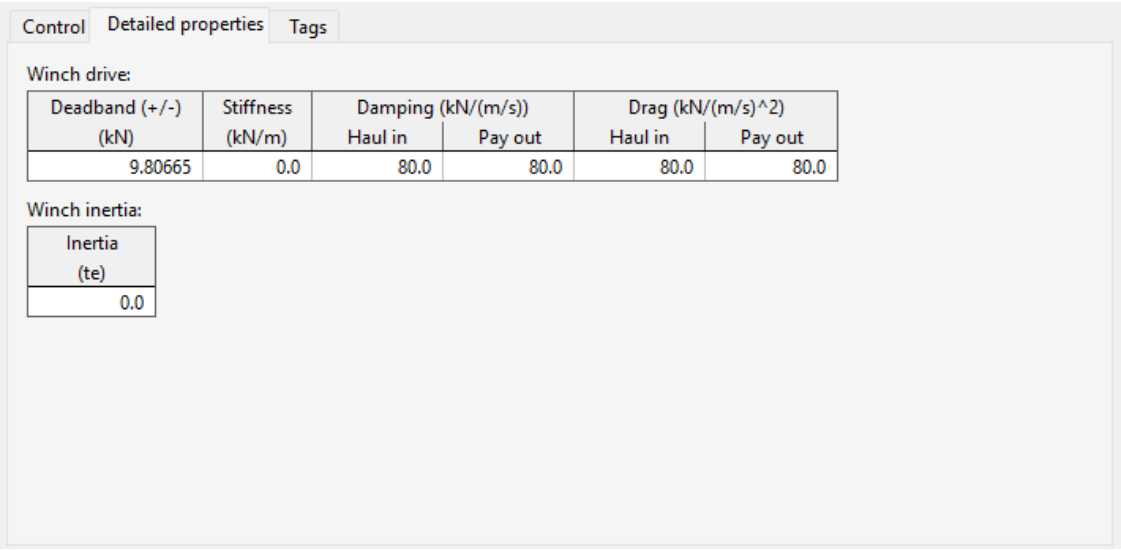
My name is Bruno Alves, I work as principal installation analysis engineer in Brazil for TechnipFMC.

I would like to model a constant tension winch for tugger winches, but I do not know how to start. The data I have available:

- Target tension: 5t;
- Max pay-out/haul-in winch speed: +/- 40m/min.

I tried to use “Detailed properties” in winch model, but it is hard to figure out the required inputs, based on the data I have for the tugger winches. Below, you can find how I set up the model:





In the case of constant tension, when it reaches the target tension plus a tolerance (something like 1t), the winch will pay-out to reduce the tension. If it reaches the target tension minus the same tolerance, the winch will haul-in to increase the tension. The winch speed shall be between 40m/min and -40m/min.

Could you please help me on this model?

Thanks and regards,

Bruno Alves

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